

ROSARIO, Argentina

WMO: 874800

Lat: 32.9081S

Lon: 60.7822W

Elev: 82

StdP: 14.65

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.3	34.0	23.2	17.6	44.5	26.6	20.6	44.5	24.4	57.0	22.3	58.1	2.8	180	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.7	94.2	73.9	91.6	73.1	89.5	72.5	78.5	88.3	76.8	86.2	75.3	84.5	10.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.6	134.2	83.7	73.8	126.4	82.1	72.3	120.1	80.5	42.1	88.0	40.4	86.2	38.9	84.6	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.7	19.5	17.1	DB	25.6	99.5	2.5	2.4	23.8	101.2	22.3	102.6	20.9	104.0	19.0	105.7
			WB	24.4	82.6	2.4	2.3	22.7	84.2	21.3	85.6	20.0	86.8	18.3	88.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	76.6	74.2	70.7	64.4	58.4	52.8	51.7	54.6	59.1	64.7	70.1	74.5	(6)
	DBStd	10.66	5.02	5.53	5.83	6.62	6.84	6.87	7.34	7.72	7.04	6.15	5.62	5.43	(7)
	HDD50	157	0	0	0	1	7	42	64	36	7	0	0	0	(8)
	HDD65	1764	1	3	17	88	228	371	419	336	201	80	17	3	(9)
	CDD50	5367	825	678	642	432	267	127	116	180	281	457	603	759	(10)
	CDD65	1502	361	261	193	70	23	7	7	15	26	72	170	297	(11)
	CDH74	14345	3805	2467	1637	502	115	18	37	134	269	663	1654	3044	(12)
	CDH80	5423	1668	960	531	113	15	1	4	30	71	184	577	1269	(13)
(14) Wind	WSAvg	7.5	7.2	6.8	6.5	6.6	6.3	6.8	7.4	8.0	8.9	9.3	8.6	7.9	(14)
(15) Precipitation	PrecAvg	39.6	4.3	4.6	6.1	3.8	2.0	1.3	1.2	1.4	2.2	4.2	4.1	4.8	(15)
	PrecMax	64.0	9.4	15.5	20.0	10.4	10.7	5.3	4.4	5.9	9.4	13.0	9.2	13.1	(16)
	PrecMin	26.3	0.4	0.2	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.7	1.2	(17)
	PrecStd	8.3	2.4	3.5	3.9	2.4	1.9	1.3	1.1	1.5	1.9	2.6	2.2	2.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.6	95.1	92.7	88.0	82.2	76.7	78.8	84.1	87.6	89.8	93.4	97.0	(19)
		MCWB	74.8	75.4	75.4	73.6	69.7	68.2	65.6	68.7	69.1	69.5	70.9	74.3	(20)
	2%	DB	94.6	91.8	88.7	82.9	77.1	71.2	72.3	77.4	80.7	84.7	89.7	93.2	(21)
		MCWB	74.1	75.0	73.2	69.6	67.2	64.6	63.0	64.5	65.6	67.9	69.8	72.8	(22)
	5%	DB	91.4	89.3	85.9	79.6	73.3	67.6	67.8	72.7	76.6	80.8	86.2	90.0	(23)
		MCWB	73.3	74.5	72.1	68.2	65.2	61.6	59.4	61.4	62.5	66.3	68.3	71.8	(24)
	10%	DB	88.5	86.1	82.6	76.5	69.7	64.3	64.1	68.2	72.5	77.2	82.8	87.4	(25)
		MCWB	72.5	72.8	70.3	66.8	63.0	58.2	57.0	58.4	60.2	64.3	67.0	70.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.3	80.5	79.4	76.1	71.9	69.5	67.8	70.2	71.7	74.6	75.3	79.8	(27)
		MCDB	91.1	91.0	88.5	83.6	78.7	74.7	75.0	80.5	83.7	84.0	87.2	89.9	(28)
	2%	WB	77.9	78.3	76.5	73.3	68.9	66.2	64.4	67.0	68.1	71.1	72.6	76.9	(29)
		MCDB	88.7	87.2	84.0	79.7	74.6	69.6	70.9	75.5	77.6	79.6	83.8	86.5	(30)
	5%	WB	76.3	76.3	74.0	70.9	66.8	63.3	61.8	63.9	65.1	68.8	70.7	74.8	(31)
		MCDB	86.7	84.9	81.7	76.6	71.9	66.6	66.7	70.6	72.6	77.6	81.4	84.7	(32)
	10%	WB	74.8	74.5	72.1	68.7	64.3	60.0	59.0	60.3	62.5	66.5	69.0	72.8	(33)
		MCDB	84.4	82.9	79.6	73.8	68.7	62.8	62.4	65.8	69.9	74.4	79.2	82.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.7	19.3	19.6	19.1	17.8	18.4	19.6	21.5	21.5	20.1	21.2	21.0	(35)
		MCDBR	24.7	22.9	23.2	23.2	20.7	19.7	23.2	26.0	26.6	25.6	26.1	25.6	(36)
		MCWBR	9.7	9.5	10.2	11.8	11.6	12.7	13.9	14.5	13.3	12.2	11.1	10.2	(37)
	5% WB	MCDBR	20.2	18.7	18.8	17.5	17.2	14.9	19.0	20.9	21.0	21.0	22.1	20.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.406	0.397	0.373	0.378	0.358	0.354	0.355	0.418	0.475	0.423	0.399	0.399	(40)
		taud	2.354	2.393	2.458	2.414	2.427	2.426	2.397	2.176	2.019	2.239	2.334	2.360	(41)
	Ebn at Noon		296	292	288	267	255	248	253	249	251	281	296	299	(42)
		Edh at Noon	42	39	34	32	28	26	29	40	52	45	43	42	(43)
(44) All-Sky Solar Radiation	RadAvg		2270	1953	1664	1213	892	762	836	1119	1485	1791	2170	2306	(44)
	RadStd		127	156	150	149	81	90	89	107	123	155	130	140	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	+0.32	+1.06	+1.13	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	+0.63	+0.90	+0.97	N/A	N/A	N/A	N/A

Nomenclature: See separate page